THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Acids, Bases and Salts** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A solution has pH 3. Another solution has pH 6. Which statement is correct? 1

(a) The first is more acidic **(b)** The second is more acidic
(c) Both are equally acidic **(d)** Both are basic

2. A solid X reacts with dilute HCl giving a gas that turns lime water milky; on strong heating, X gives Na_2CO_3. X is: 1

(a) NaCl **(b)** NaOH **(c)** $NaHCO_3$ **(d)** Na_2SO_4

3. Assertion (A): Plaster of Paris should be stored in moisture-proof containers. Reason (R): It sets into a hard mass of gypsum on contact with moisture. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Gypsum of molar mass 172 g loses 27 g on conversion to plaster of Paris. The mass of plaster of Paris obtained from 172 g is: 1

(a) 126 g **(b)** 118 g **(c)** 145 g **(d)** 172 g

5. A gas turns blue litmus red and is formed in the chlor-alkali process. The gas is: 1

(a) oxygen **(b)** hydrogen **(c)** chlorine **(d)** carbon dioxide

6. Assertion (A): Anhydrous $CuSO_4$ is used as a test for water. Reason (R): Anhydrous $CuSO_4$ is blue in colour. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. The percentage of water in $CuSO_4 \cdot 5\,H_2O$ (Cu = 63.5, S = 32, O = 16, H = 1) is approximately: 1

(a) 50% **(b)** 36% **(c)** 18% **(d)** 10%

8. A student finds the pH of solutions P, Q, R and S to be 2, 5, 9 and 13. Equal volumes of which two solutions (of equal strength) mixed together would be closest to neutral? 1

(a) R and S **(b)** P and Q **(c)** P and S **(d)** Q and R

9. **Assertion (A):** Tooth decay starts when the pH of the mouth falls below 5.5. 1
Reason (R): Tooth enamel is dissolved by the bases produced by bacteria in the mouth.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Washing soda has the formula $Na_2CO_3 \cdot 5\,H_2O$. 1
Reason (R): Washing soda is obtained by recrystallisation of sodium carbonate.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the pH of rain fall below 5.6 in industrial areas? 2

12. If 5 mL of NaOH needs 20 mL of a dilute acid, how much acid is needed for 25 mL of NaOH? 2

13. Why is the pH of an aqueous solution of NH_4Cl less than 7? 2

14. Explain why a solution of baking soda is neither strongly acidic nor strongly basic. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why is baking soda better than washing soda for a cake? Give two reasons with the effect of heat. 3

16. Pure water does not conduct electricity but adding a little acid makes it conducting. Explain. 3

17. Name a compound used to make 3
(i) glass
(ii) a cast for fractured bones and write their formulae.

OR

Calculate the percentage of water of crystallisation in gypsum (Ca = 40, S = 32, O = 16, H = 1). 3

18. A reaction mixture of $NaOH$ and HCl becomes warm. Explain, with an equation and the nature of reaction. 3

OR

Explain how a decrease in pH of a river water affects aquatic life. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a chemical factory, a concentrated solution of sodium chloride (brine) is electrolysed. Three useful substances are formed: gas X at the anode, gas Y at the cathode and a solution Z near the cathode. The gas X is used to prepare bleaching powder. 5
(i) Name the process.
(ii) Identify X and Y.
(iii) Name the solution Z.
(iv) Write the balanced equation for the electrolysis and the equation for the formation of bleaching powder from X.

20. Explain, with equations, the conversion NaCl -> NaHCO3 -> Na_2CO_3 -> $Na_2CO_3 \cdot 10\,H_2O$. Give a use for each. 5

OR

Explain, with reasoning, the colour of litmus, phenolphthalein and methyl orange in solutions of HCl, NaOH and NaCl. 5

END OF QUESTION PAPER — Total: 40 Marks